

```

*****
*
*          STAAD.Pro V8i SELECTseries3          *
*          Version  20.07.08.20                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    MAY  1, 2013                 *
*          Time=    15:49:37                    *
*
*          USER ID: Winstein Engineering Consultants  *
*****

```

1. STAAD SPACE

INPUT FILE: Pipe Access hanging scaffold 1.STD

2. START JOB INFORMATION

3. ENGINEER DATE 29-APR-13

4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

```

8. 1 0 0 0; 2 0.63 0 0; 3 2.459 0 0; 4 3.089 0 0; 5 0 1.9 0; 6 0.63 1.9 0
9. 7 2.459 1.9 0; 8 3.089 1.9 0; 9 0 3.8 0; 10 0.63 3.8 0; 11 2.459 3.8 0
10. 12 3.089 3.8 0; 13 0 0.5 0; 14 0.63 0.5 0; 15 0 1 0; 16 0.63 1 0
11. 17 2.459 2.4 0; 18 3.089 2.4 0; 19 2.459 2.9 0; 20 3.089 2.9 0; 21 0 0 2.2
12. 22 0.63 0 2.2; 23 2.459 0 2.2; 24 3.089 0 2.2; 25 0 1.9 2.2; 26 0.63 1.9 2.2
13. 27 2.459 1.9 2.2; 28 3.089 1.9 2.2; 29 0 3.8 2.2; 30 0.63 3.8 2.2
14. 31 2.459 3.8 2.2; 32 3.089 3.8 2.2; 33 0 0.5 2.2; 34 0.63 0.5 2.2; 35 0 1 2.2
15. 36 0.63 1 2.2; 37 2.459 2.4 2.2; 38 3.089 2.4 2.2; 39 2.459 2.9 2.2
16. 40 3.089 2.9 2.2; 41 0 0 1.1; 42 0.63 0 1.1; 43 2.459 0 1.1; 44 3.089 0 1.1
17. 45 0 1.9 1.1; 46 0.63 1.9 1.1; 47 2.459 1.9 1.1; 48 3.089 1.9 1.1
18. 49 0 3.8 1.1; 50 0.63 3.8 1.1; 51 2.459 3.8 1.1; 52 3.089 3.8 1.1
19. 53 0 0.5 1.1; 54 0.63 0.5 1.1; 55 0 1 1.1; 56 0.63 1 1.1; 57 2.459 2.4 1.1
20. 58 3.089 2.4 1.1; 59 2.459 2.9 1.1; 60 3.089 2.9 1.1; 61 0 4.3 0
21. 62 0.63 4.3 0; 63 0 4.3 2.2; 64 0.63 4.3 2.2; 65 0 4.3 1.1; 66 0.63 4.3 1.1
22. 67 0 4.8 0; 68 0.63 4.8 0; 69 0 4.8 2.2; 70 0.63 4.8 2.2; 71 0 4.8 1.1
23. 72 0.63 4.8 1.1; 73 0 2.4 0; 74 0 -0.2 0; 75 0.63 -0.2 0; 76 2.459 -0.2 0
24. 77 3.089 -0.2 0; 78 0 -0.2 2.2; 79 0.63 -0.2 2.2; 80 2.459 -0.2 2.2
25. 81 3.089 -0.2 2.2; 82 -0.4 -0.2 0; 83 -0.4 -0.2 2.2; 84 3.489 -0.2 0
26. 85 3.489 -0.2 2.2; 86 -0.4 -0.2 -0.15; 87 3.489 -0.2 -0.15; 88 -0.4 -0.2 2.35
27. 89 3.489 -0.2 2.35; 90 -0.4 -0.2 1.1; 91 3.489 -0.2 1.1; 92 -0.4 -0.564 -0.15
28. 93 3.489 -0.564 -0.15; 94 -0.4 -0.382 1.1; 95 3.489 -0.382 1.1
29. 96 -0.4 2.8 -0.15; 97 3.489 2.8 -0.15

```

30. MEMBER INCIDENCES

```

31. 1 1 2; 2 2 3; 3 3 4; 4 5 6; 5 6 7; 6 7 8; 7 9 10; 8 10 11; 9 11 12; 10 1 13
32. 11 5 73; 12 2 14; 13 6 10; 14 3 7; 15 7 17; 16 4 8; 17 8 18; 18 13 15
33. 19 14 16; 20 15 5; 21 16 6; 22 17 19; 23 18 20; 24 19 11; 25 20 12; 27 21 22
34. 28 22 23; 29 23 24; 30 25 26; 31 26 27; 32 27 28; 33 29 30; 34 30 31; 35 31 32
35. 36 21 33; 37 25 29; 38 22 34; 39 26 30; 40 23 27; 41 27 37; 42 24 28; 43 28 38
36. 44 33 35; 45 34 36; 46 35 25; 47 36 26; 48 37 39; 49 38 40; 50 39 31; 51 40 32
37. 52 24 29; 53 1 12; 57 45 46; 58 46 47; 59 47 48; 60 49 50; 61 50 51; 62 51 52
38. 63 41 53; 64 45 49; 65 42 54; 66 46 50; 67 43 47; 68 47 57; 69 44 48; 70 48 58
39. 71 53 55; 72 54 56; 73 55 45; 74 56 46; 75 57 59; 76 58 60; 77 59 51; 78 60 52
40. 80 1 41; 81 41 21; 82 13 53; 83 53 33; 84 15 55; 85 55 35; 86 8 48; 87 48 28

```

STAAD SPACE

-- PAGE NO. 2

41. 88 18 58; 89 58 38; 90 20 60; 91 60 40; 92 2 42; 93 42 22; 94 4 44; 95 44 24
42. 96 3 43; 97 43 23; 98 41 42; 99 7 47; 100 47 27; 101 33 34; 102 13 14
43. 103 37 38; 104 17 18; 105 34 37; 106 36 39; 107 54 57; 108 56 59; 109 9 61
44. 110 61 67; 111 49 65; 112 65 71; 113 29 63; 114 63 69; 115 30 64; 116 64 70
45. 117 50 66; 118 66 72; 119 10 62; 120 62 68; 121 9 49; 122 49 29; 123 10 50
46. 124 50 30; 125 57 66; 126 59 72; 127 17 62; 128 19 68; 129 24 18; 130 73 9
47. 131 21 73; 132 12 52; 133 52 32; 134 61 65; 135 65 63; 136 67 71; 137 71 69
48. 138 63 64; 139 19 20; 140 39 40; 141 5 45; 142 45 25; 143 74 1; 144 78 21
49. 145 79 22; 146 80 23; 147 81 24; 148 76 3; 149 77 4; 150 75 2; 151 74 75
50. 152 75 76; 153 76 77; 154 78 79; 155 79 80; 156 80 81; 157 82 74; 158 83 78
51. 159 77 84; 160 81 85; 161 82 90; 162 84 91; 163 86 82; 164 87 84; 165 83 88
52. 166 85 89; 167 90 83; 168 91 85; 169 92 94; 170 93 95; 171 94 88; 172 95 89
53. 173 94 90; 174 95 91; 175 92 90; 176 93 91; 177 88 96; 178 89 97
54. START USER TABLE
55. TABLE 1
56. UNIT METER KN
57. ANGLE
58. UA65X65X6
59. 0.065 0.065 0.006 0.0128029 0.00026 0.00026
60. END
61. DEFINE MATERIAL START
62. ISOTROPIC STEEL
63. E 2.05E+008
64. POISSON 0.3
65. DENSITY 76.8195
66. ALPHA 1.2E-005
67. DAMP 0.03
68. TYPE STEEL
69. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
70. ISOTROPIC STAINLESSSTEEL
71. E 1.9793E+008
72. POISSON 0.3
73. DENSITY 76.8195
74. ALPHA 1.8E-005
75. DAMP 0.03
76. END DEFINE MATERIAL
77. MEMBER PROPERTY BRITISH
78. 1 TO 25 27 TO 53 57 TO 78 80 TO 150 TABLE ST PIP482.5
79. 151 TO 160 TABLE ST UC152X152X30
80. MEMBER PROPERTY BRITISH
81. 161 TO 176 UPTABLE 1 UA65X65X6
82. 177 178 PRIS YD 0.016
83. CONSTANTS
84. BETA 45 MEMB 161 TO 176
85. MATERIAL STEEL MEMB 1 TO 25 27 TO 53 57 TO 78 80 TO 176
86. MATERIAL STAINLESSSTEEL MEMB 177 178
87. SUPPORTS
88. 86 87 92 93 96 97 PINNED
89. MEMBER TENSION
90. 177 178
91. LOAD 1 LOADTYPE NONE TITLE DEAD LOAD
92. SELFWEIGHT Y -1
93. MEMBER LOAD
94. 59 60 98 UNI GY -0.22
95. 6 7 27 32 33 151 UNI GY -0.11
96. 92 93 99 100 123 124 UNI GY -0.27

STAAD SPACE

-- PAGE NO. 3

97. LOAD 2 LOADTYPE NONE TITLE LIVE LOAD
 98. MEMBER LOAD
 99. 59 98 UNI GY -2.42
 100. 6 27 32 151 UNI GY -1.21
 101. 92 93 99 100 UNI GY -1.37
 102. LOAD COMB 3 COMBINATION LOAD CASE 3
 103. 1 1.4 2 1.6
 104. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 97/ 173/ 6

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 73/ 23/ 144 DOF
 TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 564
 SIZE OF STIFFNESS MATRIX = 82 DOUBLE KILO-WORDS
 REQD/AVAIL. DISK SPACE = 13.1/ 286709.8 MB

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 96 EQN.NO. 157
 LOADS APPLIED OR DISTRIBUTED HERE FROM ELEMENTS WILL BE IGNORED.
 THIS MAY BE DUE TO ALL MEMBERS AT THIS JOINT BEING RELEASED OR
 EFFECTIVELY RELEASED IN THIS DIRECTION.
 ZERO STIFFNESS IN DIRECTION 5 AT JOINT 96 EQN.NO. 158
 ZERO STIFFNESS IN DIRECTION 6 AT JOINT 96 EQN.NO. 159
 ZERO STIFFNESS IN DIRECTION 4 AT JOINT 97 EQN.NO. 562
 ZERO STIFFNESS IN DIRECTION 5 AT JOINT 97 EQN.NO. 563
 ZERO STIFFNESS IN DIRECTION 6 AT JOINT 97 EQN.NO. 564

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

ZERO STIFFNESS IN DIRECTION 4 AT JOINT 96 EQN.NO. 157
 LOADS APPLIED OR DISTRIBUTED HERE FROM ELEMENTS WILL BE IGNORED.
 THIS MAY BE DUE TO ALL MEMBERS AT THIS JOINT BEING RELEASED OR
 EFFECTIVELY RELEASED IN THIS DIRECTION.
 ZERO STIFFNESS IN DIRECTION 5 AT JOINT 96 EQN.NO. 158
 ZERO STIFFNESS IN DIRECTION 6 AT JOINT 96 EQN.NO. 159
 ZERO STIFFNESS IN DIRECTION 4 AT JOINT 97 EQN.NO. 562
 ZERO STIFFNESS IN DIRECTION 5 AT JOINT 97 EQN.NO. 563
 ZERO STIFFNESS IN DIRECTION 6 AT JOINT 97 EQN.NO. 564

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

105. PARAMETER 1
 106. CODE BS5950
 107. LY 2.5 MEMB 161 TO 168
 108. LY 2.53 MEMB 169 TO 172